

A Participatory Game Design Process for Learning Hamiltonian Path Problem

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Abstract

Hamiltonian path problem is an important problem in algorithm and graph theories. The concept and illustration of this problem cannot easily described and understand in the regular textbook. To enhance understanding this problem and developing an intriguing description of this problem, we proposed a participatory game design process to instruct the college students to develop a set of games to describe the Hamiltonian path problem. In the process, the students are arranged and instructed to design a set of games with the Hamiltonian path. Then, the students are guided to develop the app in ipad with these games.

This paper presents each stage of the proposed process. The design of process is our research of exploring methods to describing the Hamiltonian path problem. We also instruct a group of students to develop a game app following the process. At the end, we will discuss our findings.

1 Introduction

The Hamiltonian path problem is an important problem and can be applied to many different applications [4][9]. To describe this problem for the general readers, a popular science book with many vivid descriptions for learning the Hamiltonian path problem is published. In the book, the Hamiltonian path problem was described as the routes and mazes in the zoo. With careful design, this book is polished with many interesting examples for general readers even the juvenile. The authors hope that more readers can understand the beauty of the Hamiltonian path problem. The young readers are all digital natives. According to the study[5], the characteristics of these digital natives are the followings:

1. The digital natives frequently use the mobile devices such as ipad or iPhone;
2. The digital native are familiar with the

computer and network technologies;

3. The digital natives frequently use web applications and mobile phone based technologies.

To make this group of readers to understand and interest this problem, the digital version of the original book should be designed and created for these digital natives. However, the presentations and scenarios of this digital version should fit the above characteristics. Therefore, the interactive games for introducing the Hamiltonian path problems could be highly popular for these digital natives. How can we archive the goal of design the best digital content for introducing the Hamiltonian path for the digital natives? Based on our analysis, an app application in tablet computer such as ipad would be popular for these young digital natives.

Of course, there is another important conjecture: the young digital native might understand what the popular items are for other young people. If we can guide the college students who are computer-related major to develop the Hamiltonian path app, the app would be more illustrative for other young people. The notion of this idea fit to the important HCI design concept, the participatory design process.

The participatory design is an approach to invite the users to join the design team of a new system design. The users involve in design process and will in charge of some parts of the system functions. In this paper, based on the notion of the participatory design, we propose the participatory game design process. In this process, instead of create an app for the young college students to learn the Hamiltonian path problem; a group of student are invited to take part as the designer and create the games using the Hamiltonian path. The teachers as the researchers will provide the guidelines and knowledge whenever the students need in the whole process. In addition, the researchers will design the steps and phases of the participatory game design process. Therefore, the

process can be reused for another design project similar to this project. As the project is completed, we will observe the whole process and report the findings for future development. The participatory game design process was used as the tool to instruct a team of students in the Department of Computer Science and Communication Engineering at Providence University to develop an app game for learning the Hamiltonian path problem. In this paper, the final and intermediary results of this project are presented. From the results, we find that the participants can learn the Hamiltonian Path problem and the technologies for designing the app.

The remaining of this paper is organized as the follows. Section 2 gives a brief description of the methodology and theorems used in our proposed process. Section 3 is the description of the participatory game design process; Section 4 shows the results that we used this process to guide the team of student to develop the app for learning the Hamiltonian path problem. Conclusions and future study are in the final section.

2 Preliminary Studies

Our design is originate with participatory design, the techniques for game design and knowledge space theory. The participatory design is a mature human computer interface (HCI) design technique [6]. The participatory design featuring techniques about how the users play toe roles in design. Several concerns in the participatory design include:

- The design politics in the design process;
- The software analysis and design tools used in the process;
- The natural of participation.

These three concerns are key factors for the success of the design. Especially, the natural of participation is a critical factor. Recently, the research studies related to the participatory design focused on this issues [1][3].

On the contrast, the game design is a very new HCI research topic. The players of the game become the main focus in the study of the game design. Therefore many strategies to find the profiles and perspective of the players have been developed [7][10]. The research of the game design focus on four aspects: learning, storytelling, game play and user experience. The game design related study showed that how to design criteria and framework for evaluating the previous four aspects are the mainstream research topics in game design. In

this paper, we will put these considerations into the design of our process.

The third methodology we use in the design is the knowledge space theory. This is the theory used for eLearning design [8] or for diagnosis for the states in learning mathematics. Usually, the knowledge is ordered into different sets. Among these sets, some are labeled as the prior knowledge sets and the others are labeled as the posterior knowledge sets. The prior knowledge sets are prerequisite before the posterior knowledge sets. Hence, the sequence of these knowledge sets will be organized as the learning schedule. The learning path can be design by using the knowledge space theory.

In this paper, we use these three methodologies in design the participatory game design process to guide the students to develop the games for learning the Hamiltonian path problem. The detailed of the process will be described in next section.

3 Participatory Game Designs

The participatory game design process shown in Figure 1 includes four stages. Each stage has its functions. The students will be guided following the stages to complete the development of app. In addition, we also hope the students can learn the important features of the Hamiltonian path problem when they are training to develop the app. In this design, based on the idea about participatory design and game design, the four stages are (1) knowledge accumulation, (2) game sketch design, (3) programming and development and (4) App game development and presentation.

In the knowledge accumulation stage, students are taught the knowledge of the Hamiltonian path problem. We used the textbook of discrete mathematics and the popular science book[4] as the teaching material to teach the students. In addition, we have created a prototype game system as showing how to design the game for learning the Hamiltonian path. The prototype system is presented as the app in the smartphone. In this stage, the students are taught to equipped with the comprehensive knowledge of the Hamiltonian path problems. If students would like to know more, the research related to Hamiltonian path[2][9] are used as the supplement showing the applications of the Hamiltonian path problem. When the stage is finished, the students are evaluated whether they understand the

Hamiltonian path problem.

The knowledge accumulation stage is the preparation stage. The participating students are taught with the knowledge, so that they can participate the design process. The prototype system is used to illustrate the goal of this project. When students are ready, they are led into the second stage, game sketch design stage.

In the game sketch design stage, students are asked to play the leading roles to design and develop the story of the game. The students are given the game design strategies. In addition, we will teach these student how to develop the game scenarios. After of these lectures, the students are required to develop the story of the game using the Hamiltonian path. At beginning, the students will draw the storyboards for the whole game. The game can have several storyboards. For each storyboard, the students will draw the sketches of each scene in this storyboard. In the scenes, the graph of the Hamiltonian path should be deployed in the scene. We review these sketches and make sure the Hamiltonian path is correctly put in these scenes. Then, the 2D animations for the game will be created by students based on these sketches. When these animations are completes, the focus will move to how to create the games using the programming tools.

In the programming and development stage, the students are trained to write the app applications. The students will learn how to write the app and how to deploy the game as an app. Several programming tools such as pair programming for agile development will be taught so that the students can develop the app in a tight schedule. The reviewing process will confirm these students with the capability to develop the app. In addition to the programming tools, the lectures of HCI design and using multimedia in app are also taught. At the final, the students are asked to setup a schedule to complete the development of the app for learning the Hamiltonian path problem.

The app game development and presentation stage is the final stage to finish the app development. The students are finish the development based on their own schedules. The weekly review by the teachers will help students finding the drawbacks and errors in the their project. At the end, the outcome of this project is a app application in iPad for the youngsters. Based on the core value of the participatory process, the youngsters are invited as the users to test the app as the game players. The final evaluation will be used to find out whether these children have learned the concepts of

Hamiltonian path.

4 Project of Design the app

Based on the framework, we had instructed a group of seniors in the Computer Science and Communication Engineering Department to develop an app for learning the Hamiltonian path. The project started at the lecture of learning the Hamiltonian path problem using the book, Mathematics in Zoo[4]. The students quickly caught the concepts and draw the storyboard to put the Hamiltonian path in a treasure hunting game. Later, based on their discussion, the storyboard of the game were changed to mazes in three different scenes. One storyboard was a water tunnel in the dessert to find water for the thirsty camel, Figure 2. Another storyboard was the maze in the grassland to save the rabbits, Figure 3. The final one based on the story of the little red hat was a maze in the forest to escape the wolf, Figure 4.

The app was developed and installed in a iPad. The students created several different graphs and different scenarios. It is successful for these students to understand the Hamiltonian path problem. In addition, the students had the capability to develop the app, designed and created games and also understand the process of the game development. When this project was completed, this group of students won the second prize in the project contest at the department of Computer Science and Communication Engineering, Providence University. Later, they will use this app to teach children the Hamiltonian path problem. It is a successful design of the pedagogical framework for software education and computer mathematics.

5 Conclusion and Future Study

In this paper, we propose a participatory game design process to teach the students to develop the games for learning the Hamiltonian path problem. The results show the proposed process is robust. In the future, we will collect more data for improve our developments. In the future, we will conduct this project in the following directions:

1. Developing more reviewing process and the guidelines for helping the students can understand the whole process;
2. Finding the interactiveness and useabilities of the results of this project, so that we can develop different learning app for the Hamiltonian path problem.

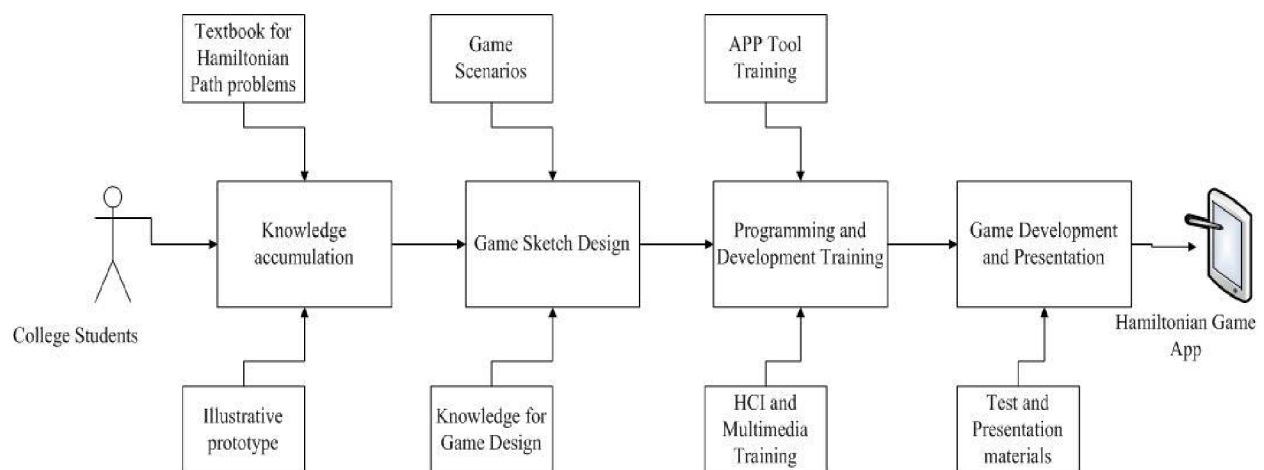


Figure 1. The Participatory Game Design Process

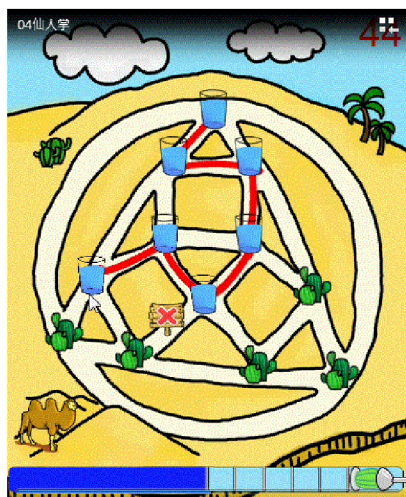


Figure 2 The first game scene

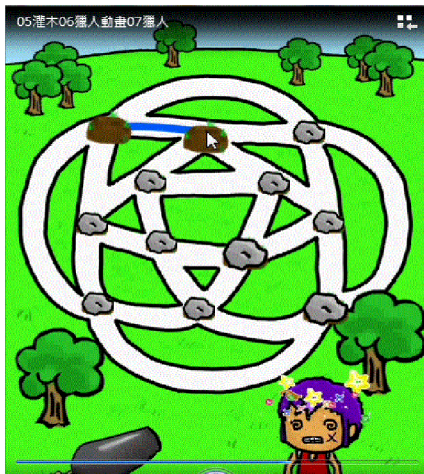


Figure 3 The second game scene



Figure 4 The third game scene

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